

Harshdeep Sharma

Computational Mechanics Engineer | FEM · Fracture · HPC · Physics-ML

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Professional Summary

Computational Mechanics Engineer with Ph.D.-level expertise in **finite element modeling, fracture and fatigue mechanics, and high-performance scientific computing**, with growing specialization in **physics-informed and hybrid data-driven solvers**. Proven experience designing **custom FEM solvers (UEL/UMAT)**, accelerating nonlinear PDE simulations, and validating numerically stable models for real engineering systems.

Strong background in **mechanics-aware SciML**, combining FEM with PINNs and operator-based learning to improve computational efficiency, robustness, and interpretability. Comfortable working across **C++ / Python / HPC environments**, translating mathematical models into scalable, production-ready simulation workflows. Recently expanded expertise into **end-to-end AI Engineering**, building high-performance Neural Operators in **JAX** and exploring **Agentic AI frameworks**.

Professional Experience

Senior Research Scientist

ZenteiQ Aitech Innovations (Airex Lab, IISc Bangalore)

Jan 2026 – Present

- Designed and implemented computationally heavy scientific machine learning (SciML) models from scratch, including **GINO, FNO, Transolver, and WNO**, using **JAX** for an advanced research library.
- Architected a modular, reusable Python codebase emphasizing efficient memory management, end-to-end execution, and system scalability.
- Evaluated **Agentic AI frameworks**, researching multi-agent architectures, RAG (Retrieval-Augmented Generation) operations, and system observability for production-grade AI workflows.

Selected Engineering & Computational Contributions

- Designed and implemented accelerated phase-field fracture solvers for heterogeneous and bimaterial systems, achieving **2.33× reduction in computational runtime**, enabling large-scale parametric and fatigue simulations on HPC clusters.
- Developed custom **Abaqus UEL/UMAT** subroutines for high-cycle fatigue and mixed-mode fracture in composite and adhesively bonded structures, ensuring numerical stability under long-cycle loading.
- Built physics-consistent damage accumulation frameworks integrating fracture mechanics with fatigue evolution laws, validated against benchmark problems and experimental trends.
- Implemented hybrid **FEM–PINN formulations** for fourth-order phase-field fracture models, enforcing PDE constraints to improve solution regularity and generalization across material gradients.
- Open-sourced scalable FEM and fatigue simulation codes, designed for modular extension, matrix-splitting strategies, and parallel execution.
- Applied algorithmic optimization and data-structure-driven reasoning to numerical solvers, reducing memory overhead and improving nonlinear convergence behavior.

Education

Ph.D. in Computational Mechanics (Thesis Submitted)

Indian Institute of Technology Patna, India

CGPA: 9.5 / 10

Focus: Fracture, fatigue, nonlinear FEM, and physics-informed modeling of heterogeneous and interface-dominated materials

- Developed high-fidelity FEM solvers using phase-field and cohesive zone approaches.
- Implemented and validated multiple custom material models via UEL/UMAT.
- Emphasized numerical robustness, error awareness, and scalability for long-horizon simulations.

M.Tech. in Mechanical System Design (2018–2020)

Indian Institute of Technology Indore, India

CGPA: 8.82 / 10

Master's Project: Crystal Plasticity FEM for FCC materials

- Implemented physics-based CPFEM UMAT for aluminum alloys across strain rates from 10^{-3} to 10^3 s^{-1} .
- Achieved improved agreement with experimental data compared to baseline literature models.
- Released open-source implementation with community adoption on GitHub.

B.Tech. in Mechanical Engineering (2013–2017)

UIET, Kurukshetra University, India

Undergraduate Project: HCF/LCF life prediction using S–N and ϵ –N based numerical models

Technical Skills

Computational Mechanics & Numerics:

Finite Element Analysis, Nonlinear PDEs, Fracture & Fatigue Modeling, Phase-Field Methods, Cohesive Zone Models, Structural Reliability, Multiscale Modeling

Programming & Scientific Computing:

C++, Python (JAX, PyTorch, TensorFlow, NumPy, SciPy), Fortran, MATLAB; HPC workflows, solver profiling, System Design concepts

Simulation & CAE Tools:

Abaqus (UEL, UMAT), ANSYS APDL, COMSOL, FEniCS, Docker, Gmsh, ParaView

Physics-ML, SciML & AI Engineering:

Neural Operators (FNO, GINO, WNO, Transolver), Physics-Informed Neural Networks (PINNs), hybrid FEM–ML solvers, Agentic AI, RAG

Selected Technical Publications

- **Composites Part A: Applied Science and Manufacturing (2025):** Degradation-informed load accumulation framework for phase-field modeling of high-cycle fatigue. doi:10.1016/j.compositesa.2025.109377
- **Engineering with Computers (2025):** Combined phase-field and cohesive zone modeling for mixed-mode fracture. doi:10.1007/s00366-025-02134-y
- **International Journal of Fatigue (2024):** Numerical modeling of HCF behavior in adhesively bonded composite laminates under mixed-mode loading. doi:10.1016/j.ijfatigue.2023.108128
- **International Journal of Mechanics and Materials in Design (2024):** Efficient phase-field solver for elastic–plastic fracture in bimetals. doi:10.1007/s10999-023-09665-6

Full publication list available upon request.

Additional Technical Experience

Algorithmic Problem Solving & Systems Thinking

- Solved 300+ problems on Codeforces (Specialist rating: 1422), applying advanced data structures, graph algorithms, and optimization under time and memory constraints.
- Translated algorithmic efficiency principles into numerical solver optimization strategies.

FEM Development with deal.II

- Implemented 2D elasticity and Poisson solvers in C++, including mesh handling, assembly pipelines, solvers, and post-processing.

Certifications

Kaggle Micro-Certifications: Python, Pandas, Data Visualization, Machine Learning, Deep Learning

Availability

Ph.D. thesis submitted and reviewed; finalizing defense. Available to join with a 2-month notice period.